BERTINI: THE GENERAL MEMBER OF A FAMILY OF DIVISORS CUT BY A GENERATING SYSTEM ON A SMOOTH PROJECTIVE IMMERSION IS SMOOTH

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file varieties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof contains commutative diagrams typeset with Xy-pic; the retained chapter preamble supplies the native Xy-pic setup, so they compile exactly as in the source.

Editorial note. The counted span cites ten cross-chapter labels, each recorded in assembly.external_references with its canonical tag as a navigation pointer.

Editorial note. The proof cites no bibliography item inside the counted span.

Allowed standard prerequisites (editorial).

- The smooth locus of a morphism is open and fibres of smooth morphisms are smooth. Stacks Project, Morphisms, Lemma morphisms-lemma-smooth-locus-open (tag 01V9) and the definition of a smooth morphism. Makes the singular locus closed and reduces the claim to a statement about its image.
- Dimension bounds for the singular locus of a family of divisors. Stacks Project, Varieties, section on Bertini theorems (the auxiliary dimension results used inside the proof). Supplies the estimate dim(Sigma) $<$ r that finishes the argument.

Source reference. Original reference 0G47: sources/stacks-chow/tags.txt, source lines 20117–20117. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original lemma 0B2L: sources/stacks-chow/tags.txt, source lines 13934–13934. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 02G1: sources/stacks-chow/tags.txt, source lines 3212–3212. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00TR: sources/stacks-chow/tags.txt, source lines 1085–1085. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 062Y: sources/stacks-chow/tags.txt, source lines 7812–7812. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Date: stacks-project:varieties.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original reference 01V9: sources/stacks-chow/tags.txt, source lines 2438–2438. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00NQ: sources/stacks-chow/tags.txt, source lines 868–868. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 00TV: sources/stacks-chow/tags.txt, source lines 1089–1089. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *In Situation 0G47 assume*

(1) *X is smooth over k,*
(2) *the image of $\psi : V \to \Gamma(X, \mathcal{L})$ generates $\mathcal{L}$,*
(3) *the corresponding morphism $\varphi_{\mathcal{L},\psi} : X \to \mathbf{P}(V)$ is an immersion.*

Then for general $v \in V \otimes_k k'$ the scheme H_v is smooth over k'.

Proof. (We observe that X is separated and finite type as a locally closed subscheme of a projective space.) Let us use the notation introduced above the statement of the lemma. We consider the projections

$$\begin{array}{ccccc} \mathbf{A}^r_k \times_k X & \longleftarrow & H_{univ} & \longrightarrow & \mathbf{A}^r_k \times_k X \\ \downarrow & \swarrow^{p} & & \searrow^{q} & \downarrow \\ X & & & & \mathbf{A}^r_k \end{array}$$

Let $\Sigma \subset H_{univ}$ be the singular locus of the morphsm $q : H_{univ} \to \mathbf{A}^r_k$, i.e., the set of points where q is not smooth. Then Σ is closed because the smooth locus of a morphism is open by definition. Since the fibre of a smooth morphism is smooth, it suffices to prove $q(\Sigma)$ is contained in a proper closed subset of $\mathbf{A}^r_k$. Since Σ (with reduced induced scheme structure) is a finite type scheme over k it suffices to prove $\dim(\Sigma) < r$ This follows from Lemma 0B2L. Since dimensions aren't changed by replacing k by a bigger field (Morphisms, Lemma 02FY), we may and do assume k is algebraically closed. By dimension theory (Lemma 0B2L), it suffices to prove that for $x \in X \setminus Z$ closed we have $p^{-1}(\{x\}) \cap \Sigma$ has dimension $< r - \dim(X')$ where X' is the unique irreducible component of X containing x. As X is smooth over k and x is a closed point we have $\dim(X') = \dim \mathfrak{m}_x/\mathfrak{m}_x^2$ (Morphisms, Lemma 02G1 and Algebra, Lemma 00TR). Thus we win if

$$\dim p^{-1}(x) \cap \Sigma < r - \dim \mathfrak{m}_x/\mathfrak{m}_x^2$$

for all $x \in X$ closed.

Since V globally generated $\mathcal{L}$, for every irreducible component X' of X there is a nonempty Zariski open of $\mathbf{A}^r$ such that the fibres of q over this open do not contain X'. (For example, if $x' \in X'$ is a closed point, then we can take the open corresponding to those vectors $v \in V$ such that $\psi(v)$ does not vanish at x'. This open will be the complement of a hyperplane in $\mathbf{A}^r_k$.) Let $U \subset \mathbf{A}^r$ be the (nonempty) intersection of these opens. Then the fibres of $q^{-1}(U) \to U$ are effective Cartier divisors on the fibres of $U \times_k X \to U$ (because a nonvanishing section of an invertible module on an integral scheme is a regular section). Hence the morphism $q^{-1}(U) \to U$ is flat by Divisors, Lemma 062Y. Thus for $x \in X$ closed and $v \in V = \mathbf{A}^r_k(k)$, if $(x, v) \in H_{univ}$, i.e., if $x \in H_v$ then q is smooth at (x, v) if and only if the fibre H_v is smooth at x, see Morphisms, Lemma 01V9.

Consider the image $\psi(v)_x$ in the stalk $\mathcal{L}_x$ of the section corresponding to $v \in V$. We have

$$x \in H_v \Leftrightarrow \psi(v)_x \in \mathfrak{m}_x \mathcal{L}_x$$

If this is true, then we have

$$H_v \text{ singular at } x \Leftrightarrow \psi(v)_x \in \mathfrak{m}_x^2 \mathcal{L}_x$$

Namely, $\psi(v)_x$ is not contained in $\mathfrak{m}_x^2 \mathcal{L}_x \Leftrightarrow$ the local equation for $H_v \subset X$ at x is not contained in $\mathfrak{m}_x^2 \Leftrightarrow \mathcal{O}_{H_v,x}$ is regular (Algebra, Lemma 00NQ) $\Leftrightarrow H_v$ is smooth at x over k (Algebra, Lemma 00TV). We conclude that the closed points of $p^{-1}(x) \cap \Sigma$ correspond to those $v \in V$ such that $\psi(v)_x \in \mathfrak{m}_x^2 \mathcal{L}_x$. However, as $\varphi_{\mathcal{L},\psi}$ is an immersion the map

$$V \longrightarrow \mathcal{L}_x / \mathfrak{m}_x^2 \mathcal{L}_x$$

is surjective (small detail omitted). By the above, the closed points of the locus $p^{-1}(x) \cap \Sigma$ viewed as a subspace of V is the kernel of this map and hence has dimension $r - \dim \mathfrak{m}_x/\mathfrak{m}_x^2 - 1$ as desired. □